

Exchanging Genomics Reports Between Pathology Labs and Medical Centers Using the mCODE FHIR Implementation Guide

BACKGROUND

The automated transfer of discrete molecular data into the EHR facilitates clinical decision making. Existing EHR data standards lack treatment insight fields (medications, trial eligibility). mCODE defines a consensus oncology data standard supporting genomic data and lightweight FHIR transmission. Pilots explored exchange between Tempus (pathology lab) and Vanderbilt/CHLA (medical centers).

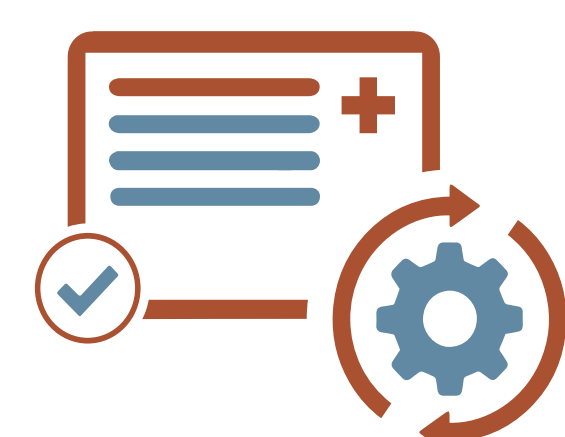
METHODS



Validation of mCODE FHIR genomic elements performed in CodeX accelerator (2021–2022).



Elements included: lab facility, order info, pathologist, demographics, specimen, molecular studies, therapeutic implications.

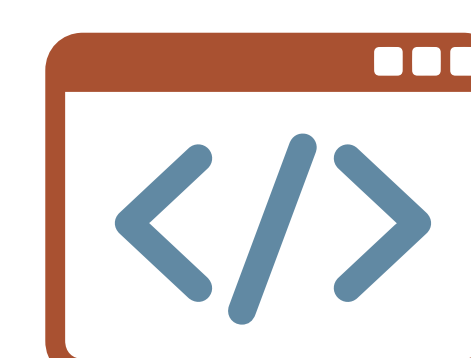


Synthetic tumor reports generated by Tempus were transformed into mCODE FHIR and transferred to academic sites.

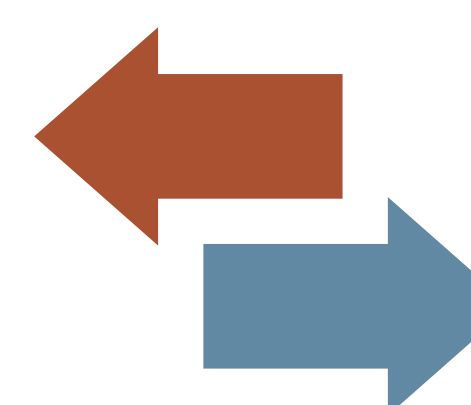
RESULTS



Over 50 Tempus genomic data elements mapped directly to mCODE



Some required added genomic annotations or syntax adjustments (eg, drug class concatenation)



mCODE files were transmitted to Vanderbilt and CHLA for decoding and downstream use



Pilots informed HL7 CGWG refinements for broader genomic compatibility

OBJECTIVE

Evaluate feasibility of mapping and exchanging molecular pathology reports into mCODE-conformant FHIR data for clinical use and standards refinement.

FUTURE DIRECTION

- Expand testing with additional labs and cancer centers
- Incorporate real-world (non-synthetic) genomic data exchange
- Continue HL7/CodeX refinements for standardization and adoption

CONCLUSION

- mCODE supports practical genomic data exchange aligned with HL7 standards
- Mapping required only nominal changes to existing pathology reports
- Collaboration between pathology labs and academic centers is critical for adoption and cross-domain interoperability